

Introduction to Exploratory Data Analysis

Course 6. Dimensionality reduction

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Principal Component Analysis (PCA)

The **principal directions** are the eigenvectors u_j of the covariance matrix $X^\top X$, unde $j \in [1, d]$.

PCA transforms the data X in $T = XU$.

Sample x_i will be transformed into the new sample t_i (line i in T).

The **principal components** are the projections of the samples on the principal directions.

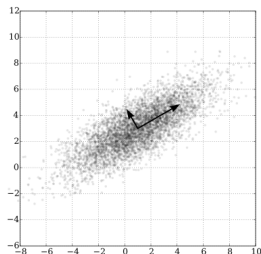
A principal component j is the column j in T .

Components are sorted descending w.r.t. the amount of variance they explain.

Inverse transformation: $X = TU^\top$

Sample x_i becomes

$$\mathbf{x}_i = \sum_{j=1}^d t_{i,j} \mathbf{u}_j^\top \quad (1)$$



PCA

Dimensionality reduction:

If instead of d (in the sum in 1) we choose to represent sample $\mathbf{x}_i$ with only $k < d$ principal components, we obtain a representation in a lower-dimensional space, i.e. dimensionality reduction.

What is left out are $d - k$ directions with low variance.

OBS1: the principal directions are not features in the original data.

OBS2: we do not have a physical interpretation on what the principal directions represent

By reducing the dimensions to 3, PCA can be used for visualizing the dataset.

By the way

PCA can be used to detect anomalies:

Since we are leaving out some of the variance when reducing sample dimensions and since the anomalies are the ones producing the most variance, reconstructing the samples with PCA will lead to lower accuracy for the anomalies.

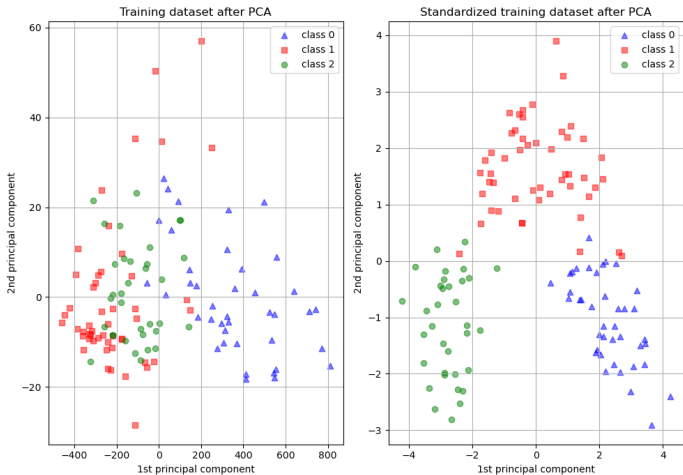
$$\mathbf{t}_i = \sum_{j=1}^k x_{i,j} \mathbf{u}_j \quad (2)$$

Reconstruction error for signal i :

$$err_i = \mathbf{x}_i - \mathbf{t}_i \quad (3)$$

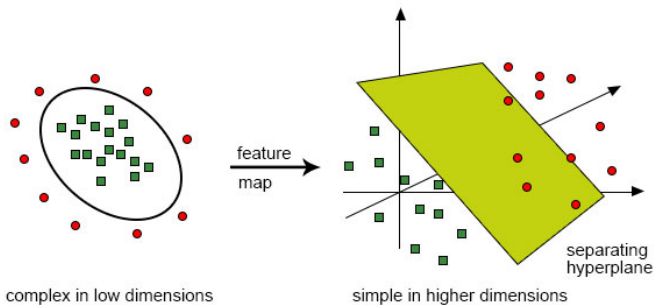
When labeling anomalies, we can use a threshold, ϵ , for deciding which samples are normal: `0 if err[i]<epsilon else 1`.

PCA and scaling



Non-linear PCA

Separation may be easier in higher dimensions



Multidimensional scaling

All points $\mathbf{x}_i$ in the dataset will have a corresponding $\mathbf{y}_i$ in a lower dimensional space.

Idea: Preserve local distances from high to low dimensional spaces, i.e.

$$\sum_{i < j} (\|\mathbf{x}_i - \mathbf{x}_j\| - \|\mathbf{y}_i - \mathbf{y}_j\|)^2 \quad (4)$$

Problem: Remember that in high dimensional spaces, samples are almost equidistant, therefore Euclidean distances don't help much.

t-SNE

t-Distributed Stochastic Neighbor Embedding

Aim: Obtain a lower-dimensional representation of the samples that preserves local structure.

Idea: Replace the distances between samples with 'similarities'.

Sample x_i is similar to x_j if the probability of x_j being the neighbor of x_i is high.

Solution:

Transform x_i in y_i (embed x_i in a lower dimensional space) such that the similarity between any pair of points x_i and x_j in the original space is equal to the similarity of their counterparts y_i and y_j .

t-SNE

For each sample $\mathbf{x}_i$ consider a Gaussian distribution centered around $\mathbf{x}_i$ (i.e. having the mean in $\mathbf{x}_i$) and with standard deviation σ_i proportional to the density around $\mathbf{x}_i$.

$$p(j|i) = \frac{\exp(-\|\mathbf{x}_i - \mathbf{x}_j\|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-\|\mathbf{x}_i - \mathbf{x}_k\|^2 / 2\sigma_i^2)} \quad (5)$$

Remember *Local Outlier Factor*, where the density around a sample was described in terms of the radius of the k-nearest neighborhood.

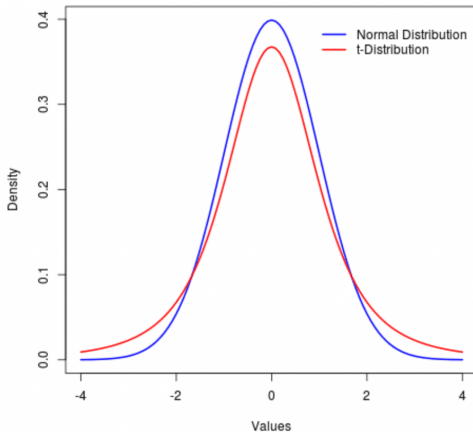
In t-SNE the density is described in terms of the variance of the Gaussian distribution: dense regions require small σ .

Number of neighbors - *perplexity* heuristic (Shannon entropy):

$$\mathcal{P} = 2^{\mathcal{H}}, \text{ where } \mathcal{H} = - \sum_{j \neq i} p(j|i) \log p(j|i) \quad (6)$$

t-SNE

In the low dimensional space - use Student-t distribution instead of Gaussian distribution.



t-SNE

Motivation: There is no space in the lower dimensional space to fit all the neighbors that exist in the high dimensional space (the *crowding problem*).

Student-t has heavier tails, i.e. allows for more distancing of neighbors that in the original space.

$$q(j|i) = \frac{(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^{-1}}{\sum_{k \neq l} (1 + \|\mathbf{y}_k - \mathbf{y}_l\|^2)^{-1}} \quad (7)$$

t-SNE

t-SNE - minimize the sum of Kullback-Leibler distances over all samples $\sum_i D_{KL}(P_i || Q_i)$, where P_i and Q_i are the probabilities in the high and low dimensional spaces respectively.

Use gradient/stochastic descent to minimize this cost.

Two components: attract neighbors and repulse distance samples.

t-SNE

Heuristics

Perplexity - controls the importance of local vs global structure

Exaggeration - controls the attraction and repulsion components

Learning rate - for the gradient/stochastic descent

See [here](#) how the choice of hyperparameters influence results (and interpretation!).

Optimization

Symmetrize the Kullback-Leibler divergence

Initialize t-SNE with PCA

Only few pairs have large dissimilarity, no need to compute all pairwise distances